Empirical Evaluation of Evolutionary NAS with Training-Free Metrics for Discovering Robust Networks under Adversarial Training

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2025 RIVF International Conference on Computing and Communication Technologies (RIVF) | 979-8-3315-7790-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/RIVF68649.2025.11365115

***Abstract*—Using training-free (TF) metrics as proxies for network performance is an effective and efficient approach in neural architecture search (NAS). Recent studies have shown that TF metrics can outperform training-based metrics in identifying robust networks when integrated as search objectives in both single-objective and multi-objective evolutionary NAS (ENAS) methods. However, prior work has typically evaluated robustness without applying adversarial training. While adversarial training can improve the robustness of deep neural networks, it has the drawback of reducing performance on clean data. In this paper, we investigate the effectiveness of ENAS with two types of performance metrics (i.e., training-based and training-free) in discovering networks that simultaneously achieve high accuracy on both clean and adversarial data. Extensive experiments on NAS problems built upon the newly proposed adversarial NAS benchmark (i.e., NAS-RobBench-201) demonstrate that multi-objective ENAS with TF metrics as search objectives can efficiently identify networks with high performance on clean data and strong robustness against various adversarial attacks.**

***Index Terms*—neural architecture search, training-free metrics, adversarial robustness, multi-objective**

I. Introduction

Deep neural networks (DNNs) plays a vital role in the impressive performance of vision models on many real-world applications [1]. However, many studies have indicated the highly vulnerable of DNNs against adversarial examples, which are the inputs but have been added small and imperceptible perturbations to cause significant misclassifications [2], [3]. The robustness against adversarial examples is thus a critical aspect when we design neural networks.

Neural architecture search (NAS) has emerged as a powerful paradigm for automating the design of high-performance neural networks [4]. Given a task and a predefined search space, NAS aims to discover the architecture that maximizes performance on the target task. Among the diverse strategies, Evolutionary NAS (ENAS), which applies evolutionary algorithms to explore candidate architectures, has become one of the most widely studied approaches [5]–[7]. Despite its effectiveness, typical ENAS remains computationally expensive because it relies on fully trained performance to evaluate candidate architectures. For example, the AmoebaNet search [7] required training $20{,}000$ models for 25 epochs over seven days on 450 K40 GPUs.

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Recent NAS studies have employed *training-free* (TF) metrics, also known as zero-cost metrics, as proxies for network performance to mitigate the substantial computational overhead of training-based evaluation [8]. These metrics can be computed from the randomly initialized network weights and typically require only a single forward-backward pass on one batch of data [8]. Most NAS research has focused on using TF metrics to discover architectures with strong performance on clean datasets, while largely overlooking their potential for identifying robust architectures. Phan *et al.* [9] evaluated a suite of TF metrics on the NAS-Bench-Suite-Zero [10] and RobustnessDataset [11] benchmarks, showing that the TF metric $SynFlow$ [12] could identify architectures with stronger robustness compared to those discovered by training-based metrics, while substantially reducing computational cost. However, their robustness evaluation was conducted *without adversarial training*. Assessing architectures after adversarial training could provide deeper insights into the effectiveness of TF metrics for discovering robust networks. It is well established that adversarial training improves robustness but often degrades performance on clean data [13]. Moreover, architectures that appear robust under standard training might not maintain high performance after adversarial training. Since models deployed in practice must balance robustness with high performance on clean inputs, jointly optimizing both objectives is critical. In this paper, we investigate the integration of zero-cost proxies into ENAS to search for architectures that not only achieve robustness under adversarial training but also preserve competitive performance on clean data.

The remainder of this paper is structured as follows. Section II reviews essential background on ENAS, adversarial attacks, and NAS benchmarks. We present our NAS pipeline for searching robustness networks and conduct experiments in Section III. We finally conclude the paper in Section IV.

II. Background and Related Work

A. Evolutionary Neural Architecture Search

Neural architecture search (NAS) is a technique that automates the design of high-performing deep neural network architectures and can be formulated as a single-objective optimization problem [5] or a multi-objective optimization problem [6], [14]. For single-objective NAS (SONAS) [5],

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the typical target is to find a single network architecture $\boldsymbol{x}^*$ within the architecture search space Ω_{arch} that has the optimal performance (e.g., the lowest classification error rate) on the validation dataset D_{val}:

$$\min_{\boldsymbol{x} \in \Omega_{\text{arch}}} f_p(\boldsymbol{x}, D_{\text{val}}). \tag{1}$$

On the other hand, multi-objective NAS (MONAS) [6], [15] seeks to discover a set of diverse networks $\mathcal{S}$ that simultaneously optimize multiple criteria beside the performance f_p:

$$\min_{x \in \Omega_{\text{arch}}} [f_p(x, D_{\text{val}}), f_2(x), \ldots, f_k(x)], \tag{2}$$

where $f_2, \ldots, f_k$ are $k-1$ objectives representing other characteristic of networks such as complexity, energy consumption, or latency. The network candidates in $\mathcal{S}$ are not Pareto dominated [16] to each other. In other words, the networks in $\mathcal{S}$ are equally good to the decision makers. A network $\boldsymbol{x^1}$ is said to Pareto dominates another network $\boldsymbol{x^2}$ (denoted as $\boldsymbol{x^1} \prec \boldsymbol{x^2}$) if and only if $\boldsymbol{x^1}$ is not worse than $\boldsymbol{x^2}$ at any objectives and $\boldsymbol{x^1}$ is better than $\boldsymbol{x^2}$ at least one objective [17]:

$$\begin{cases} \forall i \in \{1, \ldots, k\} : f_i(\boldsymbol{x^1}) \leq f_i(\boldsymbol{x^2}) \\ \exists i \in \{1, \ldots, k\} : f_i(\boldsymbol{x^1}) < f_i(\boldsymbol{x^2}) \end{cases} \tag{3}$$

Evolutionary neural architecture search (ENAS) employs evolutionary algorithms (EAs) as search optimizers to discover candidate networks within a given search space. Genetic Algorithm (GA) [18] and Non-dominated Sorting Genetic Algorithm II (NSGA-II) [16] are widely-used for solving single-objective NAS (SONAS) [5] and multi-objective NAS (MONAS) [6], respectively. Both GA and NSGA-II maintain a population of individuals (i.e., candidate networks) and progressively improve their quality over generations. At the beginning, GA and NSGA-II initialize a population P, consisting of N candidate networks randomly sampled from the search space. For each generation, crossover and mutation operators are applied to the networks in the population to produce N offspring. These offspring are then combined with the N parent networks to form a pool PO of size $2N$. A selection step is then carried out on this pool to extract N candidate networks for the next generation. This iterative process of crossover, mutation, and selection continues until either the maximum number of evaluations or the maximum number of generations is reached.

The main distinction between GA and NSGA-II lies in their selection mechanisms. In GA, the quality of each network is represented by a single metric (e.g., classification error rate), where networks with lower values (or higher values in the case of maximization) are preferred. For instance, when searching for architectures with minimal error rate, GA selects the N networks with the lowest error rates from the pool PO of $2N$ candidates. A common selection strategy is tournament selection, where the pool is partitioned into $(2N/k)$ subsets, each containing k candidates, and the best candidate from each subset is chosen to contribute to the next generation.

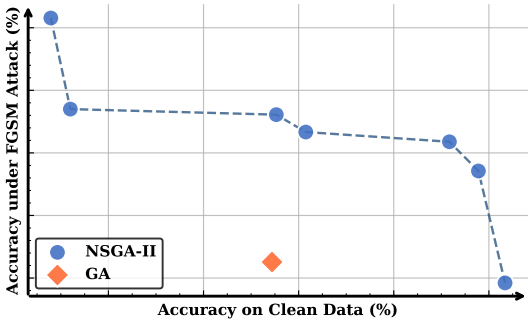

Fig. 1. The illustration presents the results of GA and NSGA-II in a bi-objective optimization problem, aiming to achieve both high accuracy on clean data and robustness accuracy.

This process is repeated until the population is filled with N candidates. In contrast, NSGA-II evaluates each candidate network across multiple objectives. Pairwise comparisons are made using the concept of Pareto dominance [16]. During the selection step, NSGA-II performs non-dominated sorting to classify solutions into different ranks (starting with rank 0). A network in rank j can only be dominated by those in lower ranks i ($\forall i < j$), meaning that rank-0 solutions are not dominated by any others. For candidates within the same rank, crowding distance [16] is calculated, with those having larger crowding distance being preferred, as they contribute greater diversity. NSGA-II progressively fills the new population by selecting solutions rank by rank until N individuals are chosen. Due to differences in their selection mechanisms, GA typically produces a single best-performing network, whereas NSGA-II yields a set of diverse, non-dominated architectures. Fig. 1 illustrates the networks discovered by GA and NSGA-II in an NAS run.

B. Synaptic Flow

Training-free (TF) NAS metrics refer to measures that can be computed without requiring the expensive and lengthy training process, and are often employed as proxies for network performance in NAS search [8], [9], [12]. Synaptic Flow ($SynFlow$) [12] is one of the widely-used TF-NAS metrics and has demonstrated effectiveness in identifying architectures with strong performance on both clean and adversarial data [9]. SynFlow is designed to measure the importance of each parameter in a network by quantifying its contribution to the loss if removed. Given a network N with parameters θ, the $SynFlow$ score for a parameter is defined as:

$$SynFlow(\theta) = \frac{\partial \mathcal{L}}{\partial \theta} \odot \theta, \tag{4}$$

where $\odot$ denotes the Hadamard product and $\mathcal{L}$ is a synthetic loss function obtained by multiplying all parameters in the network. The overall $SynFlow$ score for the network is then given by:

$$SynFlow(N) = \sum_{\theta \in N} SynFlow(\theta). \tag{5}$$

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C. Adversarial Attack and Adversarial Training

Adversarial attacks are techniques designed to cause deep neural networks (DNNs) to make incorrect predictions by introducing small, carefully crafted perturbations to the input [2] (see Fig. 2). In this paper, we experiment with white-box

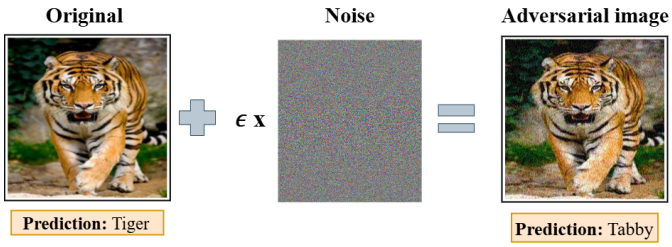

Fig. 2. Adversarial attack using FGSM [2] with $\epsilon = 0.1$. The original image (left) is correctly classified as *Tiger*, while the adversarial example (right) is misclassified as *Tabby*.

adversarial attacks, where the attacker has direct access to the internal information of the victim model, such as gradients and weights. We consider the following representative methods:

- *Fast Gradient Sign Method* (FGSM) [2] generates adversarial examples by adding a perturbation in the direction of the gradient that maximally increases the loss, thereby inducing misclassification. Given a network with the parameters θ and the loss function J, for an input x with true label y, the adversarial example x_{adv} is defined as:

$$x_{adv} = x + \epsilon \cdot sign(\nabla_x J(\theta, x, y)), \tag{6}$$

  where ϵ controls the perturbation magnitude.
- *Projected Gradient Descent* (PGD) [3] extends FGSM by applying perturbations iteratively, refining the adversarial example at each step. After every update, the perturbed input is projected back into the valid ϵ-ball around the original input to ensure the perturbation remains within the allowed range $[x - \epsilon, x + \epsilon]$. The update rule is:

$$x_{adv}^{t+1} = clip\left(x_{adv}^{t} + \alpha \cdot \text{sign}(\nabla_x J(\theta, x_{adv}^{t}, y))\right), \tag{7}$$

  where $x^0 = x$ is the original input and α is the step size.
- *AutoAttack* [19] is an ensemble framework that combines several white-box attack methods. It has been shown to generate stronger perturbations than individual attacks, achieving higher success rates in fooling victim models.

Adversarial training is a widely used approach to enhance the robustness of deep neural networks (DNNs) against adversarial attacks by training them on datasets that include both clean samples and adversarial examples [20]. While adversarial training improves robustness, it often degrades performance on clean data [13]. Identifying networks that achieve strong robustness while maintaining high performance on clean data is thus a promising research direction.

D. NAS Benchmarks

Numerous NAS studies have introduced databases known as *NAS benchmarks* [10], [21]. A NAS benchmark provides performance information for a large number of networks within the same search space, thereby enabling reproducible and efficient evaluation of NAS algorithms. NAS-Bench-201 [21] trained and evaluated 15,625 network architectures on three datasets, i.e., CIFAR-10, CIFAR-100, and ImageNet16-120, over 200 epochs. Jung *et al.* [11] introduced RobustnessDataset, which reports the robust accuracy of all NAS-Bench-201 networks under adversarial attacks such as FGSM and PGD with varying perturbation strengths. Wu *et al.* [22] further extended this by performing adversarial training on all networks in NAS-Bench-201 and evaluating their performance on both clean data and multiple adversarial attacks. The resulting information was organized into the NAS-RobBench-201 benchmark. Another important benchmark is NAS-Bench-Suite-Zero [10], which provides training-free metric (TF metric) scores for networks across various search spaces, including NAS-Bench-201. Using NAS benchmarks offers an efficient means of evaluating the effectiveness of NAS algorithms, since researchers can leverage the pre-computed information logged in these benchmarks rather than training networks from scratch. In our experiments, we query NAS-Bench-Suite-Zero for $SynFlow$ scores, NAS-Bench-201 for the performance of networks on clean datasets after standard training, and NAS-RobBench-201 for the performance of networks after adversarial training, including both their accuracy on clean data and their robustness against various adversarial attacks.

III. Experiments and Results

A. Experimental Setup

Fig. 3 illustrates the overall workflow of our experiments, which consists of two separated phases: *Search* and *Evaluation*. In the search phase, we follow the setup of [9], where four variants of GA and NSGA-II are employed to search within the NAS-Bench-201 space across three datasets (CIFAR-10, CIFAR-100, and ImageNet16-120):

- **GA (Val-Acc-12)**: uses GA as the search algorithm with the objective of maximizing the validation accuracy of networks at the 12th epoch.
- **GA (SynFlow)**: uses GA with the objective of maximizing the $SynFlow$ metric of networks.
- **NSGA-II (Val-Acc-12)**: uses NSGA-II as the search algorithm, simultaneously maximizes validation accuracy at the 12th epoch and minimizes the number of floating-point operations (lower is better).
- **NSGA-II (SynFlow)**: uses NSGA-II to jointly maximizes the $SynFlow$ score (higher is better) and minimizes the number of floating-point operations (FLOPs).

Both algorithms use the two-point crossover operator with the crossover probability of 0.9 and the integer mutation operator with a probability of $1/l$, where l denotes the length of the encoded architecture. The population size and the maximum number of evaluations are set to 20 and 3,000, respectively.

Our evaluation phase differs from [9] in that the resulting networks are additionally adversarially trained prior to

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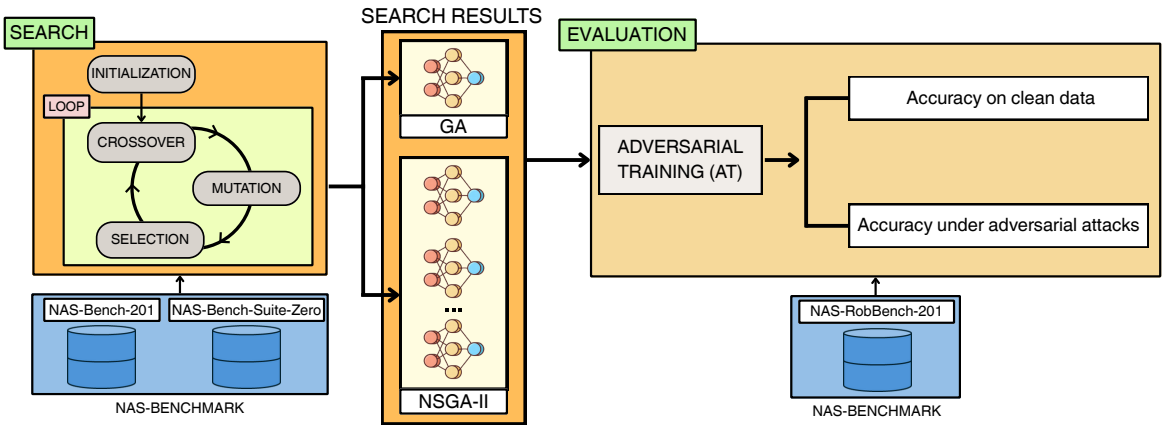

Fig. 3. An overview of the process for discovering neural architectures and evaluating their accuracy along with their robustness.

robustness assessment. Specifically, networks found by GA and NSGA-II variants undergo adversarial training and are then evaluated under two conditions: (i) no adversarial attack (representing performance on clean data), and (ii) five different adversarial attacks (representing robustness), including FGSM with $\epsilon = 3/255$, FGSM with $\epsilon = 8/255$, PGD with $\epsilon = 3/255$, PGD with $\epsilon = 8/255$, and AutoAttack. For algorithm comparison, a method A is considered better than method B if the network found by A achieves higher accuracy on both clean data and adversarial data. In the case of MONAS, where the search typically returns multiple candidate networks, we adopt the ranking mechanism from the *IEEE CEC 2024 competition on MONAS*[1] to select a single representative network for comparison with SONAS. For each kind of accuracy (i.e., under no attacks condition and under adversarial attacks), candidate networks are ranked based on their accuracy (highest accuracy assigned rank 0), and the network with the lowest total rank across all conditions is selected as the representative.

Each algorithm is executed over 31 independent runs with different random seeds, and statistical significance of the results is verified using the Student's t-test at a significance level of 0.01. Better results are shown in **bold**, while results significantly worse than the best-performing method are highlighted in gray.

B. Results of single-objective NAS

Table I reports the accuracies of networks found by GA (Val-Acc-12) and GA (SynFlow) over 31 independent runs. We observe that networks found by GA (Val-Acc-12) yield higher accuracy on both clean data ($\epsilon = 0$) and adversarially perturbed data, except under FGSM with $\epsilon = 8/255$ and PGD with $\epsilon = 8/255$ for CIFAR-10 dataset. This finding contrasts with the observations in [9], where networks identified by $SynFlow$-based GA demonstrated stronger robustness compared to those found by GA (Val-Acc-12). We note that the NAS benchmark used in our experiments (i.e., NAS-RobBench-201) differs from the one used in [9] (i.e., RobustnessDataset). While the networks in RobustnessDataset are evaluated for robustness *without* adversarial training, the networks in NAS-RobBench-201 are evaluated *after* adversarial training. We further observe that the accuracy on clean data after adversarial training for the networks with the highest validation accuracy at epoch 12th and those with the highest SynFlow score is nearly identical (77.07% vs. 77.06%, see Fig. 4). Comparing these results with Table I, we note that GA (SynFlow) consistently returned the networks with the highest $SynFlow$ score across all 31 runs, whereas GA (Val-Acc-12) did not. This phenomenon has been explained in [23]: the objective space defined by validation accuracy at epoch 12th contains many local optima, making it difficult for SONAS algorithms that optimize validation accuracy at epoch 12th to reach the global optimum (i.e., the network with the highest validation accuracy at epoch 12th). Interestingly, this difficulty inadvertently allows GA (Val-Acc-12) to discover networks with higher robustness than the one with the highest validation accuracy at epoch 12th. These findings underscore that an architecture achieving the highest performance score on clean data with standard training may not necessarily maintain its rank after adversarial training.

Moreover, under the same maximum number of evaluations (i.e., 3,000), the search cost of GA (Val-Acc-12) is substantially higher than that of training-free methods such as GA (SynFlow) runs, as it requires costly training whenever evaluating candidate networks. Specifically, the search cost of GA (Val-Acc-12) is more than $11\times$ greater than that of GA (SynFlow) [9].

C. Results of multi-objective NAS

In contrast to SONAS, the network architectures found by the training-free variant NSGA-II (SynFlow) outperform those found by NSGA-II (Val-Acc-12) across all metrics and datasets (see Table II). We suggest that solving NAS as a multi-objective problem enables the searching of multiple architectures with diverse characteristics. Such diversity appears to be beneficial for identifying networks with strong robustness against adversarial attacks. This is clearly reflected in the

[1] https://github.com/EMI-Group/IEEE-CEC-NAS-Competition

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TABLE I
AVERAGE ACCURACIES (MEAN$_{\pm\text{STD}}$) OF NETWORK ARCHITECTURES DISCOVERED BY GA VARIANTS OVER 31 INDEPENDENT RUNS.

Dataset	Search metric	Accuracy (%) ↑					
		Clean	FGSM		PGD		AutoAttack
		$\epsilon = 0$	$\varepsilon = 3/255$	$\varepsilon = 8/255$	$\varepsilon = 3/255$	$\varepsilon = 8/255$	
CF-10	Val-Acc-12	**77.70**$_{\pm 1.034}$	**68.07**$_{\pm 0.988}$	51.70$_{\pm 0.946}$	**67.55**$_{\pm 0.943}$	46.74$_{\pm 0.619}$	42.89$_{\pm 0.846}$
	SynFlow	77.06$_{\pm 0.000}$	67.64$_{\pm 0.000}$	**52.48**$_{\pm 0.000}$	67.07$_{\pm 0.000}$	**47.37**$_{\pm 0.000}$	**43.48**$_{\pm 0.000}$
CF-100	Val-Acc-12	**49.25**$_{\pm 0.642}$	**39.49**$_{\pm 0.509}$	**26.43**$_{\pm 0.388}$	**39.21**$_{\pm 0.488}$	**24.13**$_{\pm 0.306}$	**19.97**$_{\pm 0.306}$
	SynFlow	45.31$_{\pm 0.000}$	36.77$_{\pm 0.000}$	25.61$_{\pm 0.000}$	36.54$_{\pm 0.000}$	23.54$_{\pm 0.000}$	19.78$_{\pm 0.000}$
IN16-120	Val-Acc-12	**15.09**$_{\pm 0.142}$	**11.77**$_{\pm 0.046}$	**7.50**$_{\pm 0.056}$	**11.74**$_{\pm 0.054}$	**7.02**$_{\pm 0.002}$	**4.16**$_{\pm 0.069}$
	SynFlow	11.01$_{\pm 0.000}$	9.23$_{\pm 0.000}$	6.28$_{\pm 0.000}$	9.22$_{\pm 0.000}$	5.99$_{\pm 0.000}$	3.57$_{\pm 0.000}$

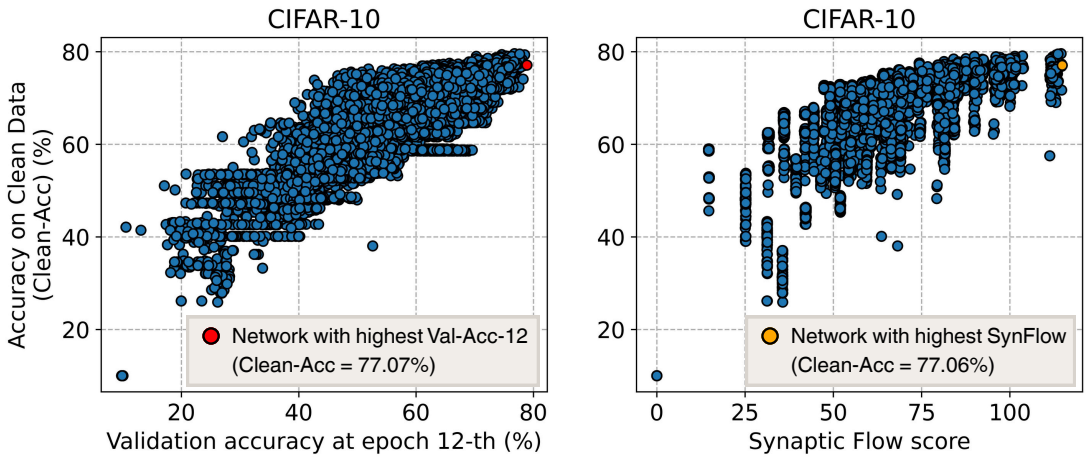

Fig. 4. Visualization of all 15,625 networks in NAS-Bench-201 with respect to (Left) validation accuracy at epoch 12th vs. accuracy on clean data after adversarial training, and (Right) $SynFlow$ vs. accuracy on clean data after adversarial training.

TABLE II
AVERAGE ACCURACIES (MEAN$_{\pm\text{STD}}$) OF NETWORK ARCHITECTURES DISCOVERED BY NSGA-II VARIANTS OVER 31 INDEPENDENT RUNS.

Dataset	Search metric	Accuracy (%) ↑					
		Clean	FGSM		PGD		AutoAttack
		$\epsilon = 0$	$\varepsilon = 3/255$	$\varepsilon = 8/255$	$\varepsilon = 3/255$	$\varepsilon = 8/255$	
CF-10	Val-Acc-12	77.36$_{\pm 0.872}$	67.70$_{\pm 0.860}$	51.42$_{\pm 0.773}$	67.20$_{\pm 0.824}$	46.57$_{\pm 0.498}$	42.67$_{\pm 0.676}$
	SynFlow	**79.37**$_{\pm 0.000}$	**69.55**$_{\pm 0.000}$	**53.51**$_{\pm 0.000}$	**68.93**$_{\pm 0.000}$	**48.11**$_{\pm 0.000}$	**44.34**$_{\pm 0.000}$
CF-100	Val-Acc-12	49.20$_{\pm 0.391}$	39.40$_{\pm 0.364}$	26.36$_{\pm 0.285}$	39.10$_{\pm 0.359}$	24.06$_{\pm 0.233}$	19.80$_{\pm 0.329}$
	SynFlow	**49.54**$_{\pm 0.000}$	**39.99**$_{\pm 0.000}$	**26.79**$_{\pm 0.000}$	**39.70**$_{\pm 0.000}$	**24.33**$_{\pm 0.000}$	**19.89**$_{\pm 0.000}$
IN16-120	Val-Acc-12	14.91$_{\pm 0.057}$	11.71$_{\pm 0.026}$	7.56$_{\pm 0.020}$	11.66$_{\pm 0.026}$	7.02$_{\pm 0.023}$	4.08$_{\pm 0.005}$
	SynFlow	**16.04**$_{\pm 0.000}$	**12.51**$_{\pm 0.000}$	**8.03**$_{\pm 0.000}$	**12.48**$_{\pm 0.000}$	**7.36**$_{\pm 0.000}$	**4.33**$_{\pm 0.000}$

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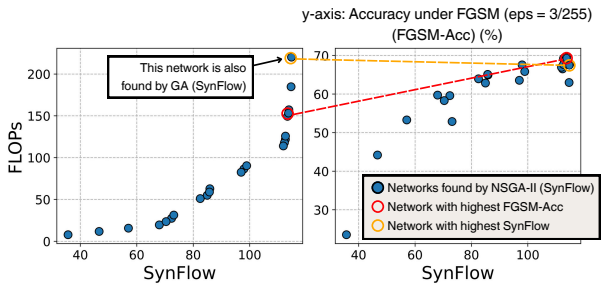

Fig. 5. The figure shows the networks discovered by NSGA-II (SynFlow) (blue points) in the median run over 31 trials. Notably, the network with the highest SynFlow score (orange circle) (this one is also the resulting network of GA (SynFlow)) does not correspond to the highest accuracy under the FGSM with $\epsilon = 3/255$ attack (red circle). On the other hand, by returning a diverse set of networks, NSGA-II (SynFlow) is able to identify architectures with higher robust accuracy than the one found by GA (SynFlow).

difference between GA (SynFlow) and NSGA-II (SynFlow). Despite both using SynFlow as the search objective, the architectures discovered by NSGA-II (SynFlow) are significantly more robust than those obtained by its single-objective variant (see Tables I and II). Fig. 5 illustrates the networks discovered by GA (SynFlow) and NSGA-II (SynFlow) in an example run. Among the four employed variants, the networks found by NSGA-II (SynFlow) achieve the highest accuracies on both clean and adversarial data. We further emphasize that the search cost of NSGA-II (SynFlow) is substantially lower than that of GA (Val-Acc-12) and NSGA-II (Val-Acc-12), demonstrating both the effectiveness and efficiency of using training-free MONAS algorithms to discover networks that achieve strong performance on clean datasets while maintaining high robustness against various adversarial attacks.

IV. Conclusions

Although adversarial training is effective for enhancing the robustness of neural networks against adversarial attacks, it often degrades performance on clean data. In this paper, we evaluate the effectiveness of evolutionary neural architecture search (ENAS) in identifying networks that achieve both high accuracy on clean data and strong robustness against various adversarial attacks. We conduct experiments under two NAS scenarios: single-objective NAS (SONAS) and multi-objective NAS (MONAS). Specifically, we employ two widely used evolutionary algorithms (i.e., genetic algorithm (GA) for SONAS, and NSGA-II for MONAS) together with two types of performance metrics: training-based and training-free. Our experiments, conducted on multiple benchmarks, yield several valuable insights. For single-objective ENAS, networks discovered using training-based metrics demonstrate better performance on both clean and adversarial data. The networks with the highest training-free metric scores do not exhibit impressive robustness. However, the training-free MONAS (TF-MONAS) variant demonstrates remarkable effectiveness. By returning a diverse set of architectures, it increases the chances of identifying networks with both high clean accuracy and strong adversarial robustness. Furthermore, using training-free metrics as search objectives is substantially more computationally efficient compared to training-based ones. While our current experiments focus only on image classification tasks and covered a limited set of search algorithms, extending the integration of TF metrics to state-of-the-art NAS strategies and validating the TF-MONAS framework in other tasks (e.g., object detection, speech recognition) are left for future work.

Acknowledgment

This research was funded by University of Information Technology - Vietnam National University HoChiMinh City under grant number D1-2025-11.

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